Seminar 3

Two routes through the consumer-theory square

Conventions. Quantities are nonnegative and $p_1, p_2, m > 0$. Targets satisfy $\bar{u} > 0$ in problem 1 and $\bar{u} > 1$ in problem 2 unless stated otherwise. Write X, H for demand correspondences and x, h when demand is unique. Use derivative identities where their derivatives exist.

Warm-up: true or false

There are exactly two goods. Preferences are continuous and locally nonsatiated; ordinary demand is unique, interior and differentiable near the point considered. Use the compensated-demand identities where defined. All classifications below are local, at the same prices and income: normal means $x_{i,m} > 0$, inferior means $x_{i,m} < 0$, and Giffen means $x_{i,p_i} > 0$. Explain each verdict briefly.

1. Both goods can be Giffen.
2. Both goods can be normal.
3. Both goods can be inferior.
4. With $p_2 = 1$, a consumer's ordinary demand is locally

$$x_1(p_1, m) = \frac{12 - m}{2 - p_1}, \qquad 1 < p_1 < \frac{4}{3}, \quad 6p_1 < m < 12.$$

   Your friend says: "Demand rises with its own price, so this cannot come from utility maximisation: it violates the law of compensated demand." Your friend is right.

5. After a small increase in p_1, with p_2 and income unchanged, the consumer buys less of good 2. Therefore good 2 is inferior.

1. Two papers, one research budget

A professor is revising two papers. Let x_i be hours of focused work on paper i. Each hour uses p_i units of a research budget m (research assistance, computing and data). The papers' quality scores are $Q_1 = \sqrt{x_1}$ and $Q_2 = \sqrt{x_2}$: more work helps, but by less and less. The second paper counts twice towards the professor's research objective, so

$$u(x_1, x_2) = \sqrt{x_1} + 2\sqrt{x_2}.$$

(a) Establish existence, uniqueness and budget exhaustion. Why must both optimal quantities be strictly positive?

(b) Solve utility maximisation directly: find $x_1(p, m)$, $x_2(p, m)$ and $v(p, m)$ for general prices and income.

(c) Solve expenditure minimisation directly: find $h_1(p, \bar{u})$, $h_2(p, \bar{u})$ and $e(p, \bar{u})$. Try $z_i = \sqrt{x_i}$.

(d) Recover x using Roy and h using Shephard. Verify

$$e(p, v(p, m)) = m, \qquad v(p, e(p, \bar{u})) = \bar{u}, \qquad h(p, v(p, m)) = x(p, m).$$

(e) At $p = (1, 4)$ and $m = 8$, find x and v. Calculate x_{1,p_1}, $x_{1,m}$ and h_{1,p_1} at the matching *fixed* target. Check the own-price Slutsky equation numerically.

(f) Compute the full Slutsky matrix from Marshallian demand:

$$S_{ij} = \frac{\partial x_i}{\partial p_j} + x_j \frac{\partial x_i}{\partial m}.$$

Check it against $D_p h$ at the matching target. Verify symmetry, $Sp = 0$ and negative semidefiniteness. Give its value at the point in (e).

(g) Generalise. For an n-good consumer with continuous LNS utility, positive prices and an attainable target above $u(0)$, suppose demands are single-valued and differentiable and e is C^2 in prices. At matching parameters, prove that S is symmetric, $Sp = 0$ and $z^\top S z \leq 0$. Use Shephard and homogeneity; derive price concavity of e from its minimisation definition.

2. Glitch: coffee, cake, and a head start

At Glitch, the NES café, a student buys coffee (x_1) and cake (x_2), measured in divisible serving-equivalents, at prices p_1 and p_2. The student has a break budget m. A familiar coffee ritual has a one-point head start; a cake break has diminishing returns. For this student, what matters is whichever of the two routes provides the better break:

$$u(x_1, x_2) = \mathsf{max}\{x_1 + 1, \sqrt{x_2}\}.$$

(a) Check continuity, local nonsatiation and convexity of preferences. Draw $\{x \geq 0 : u(x) \geq 2\}$. What is $u(0)$, and why do we work with targets $\bar{u} > 1$ for the main duality calculations?

(b) Find $v(p, m)$ and the entire $X(p, m)$ by comparing the two corner routes. At $p = (9/2, 1)$, find every income range in which each route is optimal, including both switching boundaries.

(c) Find $e(p, \bar{u})$ and the entire $H(p, \bar{u})$ for $\bar{u} > 1$. At $p = (9/2, 1)$, find every target range and both switching targets. What happens for targets $\bar{u} \leq 1$?

(d) Apply Roy and Shephard in each strict regime. Explain what prevents a single gradient from selecting a bundle at a switching boundary.

(e) Verify both inverse-value identities and demand-set equality at matching parameters. At the specified prices, match the two income boundaries to the two target boundaries.

(f) At $p = (9/2, 1)$ and $m = 9/4$, list all solutions to both corresponding problems. Is the midpoint also optimal? Check the other boundary, $m = 9$.

(g) *Extension.* Set $p = (4, 1)$ and $m = 4$. Does the derivative v_m exist at this tie? Does that alone let you use Roy's identity?